

Reflection Summary

Capstone Project 4: AI-Powered PCAP Analysis & Threat Detection

This project gave me a much better understanding of how network traffic can be investigated from a cybersecurity perspective. Before starting, PCAP files and packet-level network analysis seemed quite difficult to understand. Working with Wireshark, TShark, CSV data, Python, Excel, and Generative AI helped me understand how raw packet data can be converted into information that is easier to investigate.

One of the main things I learned is that data validation is very important. Before drawing any security conclusion, I had to confirm that the traffic data, IP addresses, ports, protocols, timestamps, and statistics were correct. I also learned that finding a large amount of traffic or repeated communication does not automatically mean a system is compromised.

Generative AI was very useful during this project because it helped me examine large amounts of traffic data and identify communication patterns, unusual ports, DNS activity, traffic bursts, and other areas that needed further investigation. However, I also learned that AI results should not be accepted without checking them. I verified important findings against the dataset and statistical analysis before using them in the final investigation.

Another important lesson for me was understanding the difference between investigation-relevant activity and confirmed malicious activity. The traffic showed several areas that required further investigation, including the communication involving 10.10.0.33 and 18.184.26.60, TCP/53480 activity, DNS traffic, and frame-size outliers. However, the available packet metadata alone was not enough to confirm malware, command-and-control communication, data exfiltration, DNS tunnelling, or compromise.

I also improved my practical skills in creating detection rules, analysing traffic statistics, creating visualizations, and developing an incident response plan. Overall, this project helped me understand how AI can make cybersecurity investigation faster, while human verification and additional evidence are still necessary to make reliable security decisions.

Final Takeaway

AI can help an analyst investigate faster, but the analyst is still responsible for validating the evidence and making the final security decision.